MSG No. 132.454 $P4'_2/m'cm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `m'.m'm'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,13,14,15,16]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,5,6,11,12,15,16]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[3,4,7,8,9,10,13,14]`

Table 3: Wyckoff site: `2c`, site symmetry: `m'.m'm'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `-42'm'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,5,6,11,12,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[3,4,7,8,9,10,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `22'2'.`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,11,12]`
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[3,4,9,10]`
3	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[5,6,15,16]`
4	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[7,8,13,14]`

Table 6: Wyckoff site: `4f`, site symmetry: `2/m'..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,13,14]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[3,4,15,16]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,9,10]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,11,12]`

Table 7: Wyckoff site: `4g`, site symmetry: `2.m'm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,15,16]`
2	$[0,\ 0,\ -z]$	`[3,4,13,14]`
3	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[7,8,9,10]`

Table 8: Wyckoff site: `4h`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,2,15,16]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[3,4,13,14]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8,9,10]`

Table 9: Wyckoff site: `4i`, site symmetry: `m'.2m'`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,3,14,16]`
2	$[-x,\ -x,\ 0]$	`[2,4,13,15]`
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[5,8,10,11]`
4	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[6,7,9,12]`

Table 10: Wyckoff site: `4j`, site symmetry: `m'.2m'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	`[1,3,14,16]`
2	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[2,4,13,15]`
3	$[x,\ -x,\ 0]$	`[5,8,10,11]`
4	$[-x,\ x,\ 0]$	`[6,7,9,12]`

Table 11: Wyckoff site: `8k`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,2]`
2	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[5,6]`
4	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[7,8]`
5	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,12]`
7	$\left[0,\ \frac{1}{2},\ -z\right]$	`[13,14]`
8	$\left[\frac{1}{2},\ 0,\ z\right]$	`[15,16]`

Table 12: Wyckoff site: `8l`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[-x,\ 0,\ \frac{1}{4}\right]$	`[2,12]`
3	$\left[0,\ x,\ \frac{3}{4}\right]$	`[3,9]`
4	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[4,10]`
5	$\left[0,\ -x,\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[0,\ x,\ \frac{1}{4}\right]$	`[6,16]`
7	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[7,13]`
8	$\left[x,\ 0,\ \frac{3}{4}\right]$	`[8,14]`

Table 13: Wyckoff site: `8m`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,11]`
2	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[2,12]`
3	$\left[\frac{1}{2},\ x,\ \frac{3}{4}\right]$	`[3,9]`
4	$\left[\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[4,10]`
5	$\left[\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	`[5,15]`
6	$\left[\frac{1}{2},\ x,\ \frac{1}{4}\right]$	`[6,16]`
7	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[7,13]`
8	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[8,14]`

Table 14: Wyckoff site: `8n`, site symmetry: `m'..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14]`
2	$[-x,\ -y,\ 0]$	`[2,13]`
3	$[y,\ x,\ 0]$	`[3,16]`
4	$[-y,\ -x,\ 0]$	`[4,15]`
5	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[5,10]`
6	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[6,9]`
7	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[7,12]`
8	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[8,11]`

Table 15: Wyckoff site: `8o`, site symmetry: `..m'`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,16]`
2	$[-x,\ -x,\ z]$	`[2,15]`
3	$[x,\ x,\ -z]$	`[3,14]`
4	$[-x,\ -x,\ -z]$	`[4,13]`
5	$\left[x,\ -x,\ \frac{1}{2}-z\right]$	`[5,11]`
6	$\left[-x,\ x,\ \frac{1}{2}-z\right]$	`[6,12]`
7	$\left[-x,\ x,\ z+\frac{1}{2}\right]$	`[7,9]`

continued ...

Table 15

No.	position	mapping
8	$\left[x,\, -x,\, z+\frac{1}{2}\right]$	[8,10]

Table 16: Wyckoff site: 16p, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-x,\, -y,\, z]$	[2]
3	$[y,\, x,\, -z]$	[3]
4	$[-y,\, -x,\, -z]$	[4]
5	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[5]
6	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[6]
7	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[7]
8	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[8]
9	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[9]
10	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[10]
11	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[11]
12	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[12]
13	$[-x,\, -y,\, -z]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$[-y,\, -x,\, z]$	[15]
16	$[y,\, x,\, z]$	[16]